

ELEC2134 : Circuits and Signals

PART A (Week 1 to Week 5)

Transform Methods

- Periodic signals and Fourier series
- Aperiodic signals and Fourier transforms
- Laplace transforms of signals and circuits

System Analysis

- Linear systems and convolution
- Impulse response
- Frequency response and transfer functions
- Steady-state analysis of linear systems
- Transient analysis of 1st and 2nd systems
- Circuit analysis using transform methods

PART B (Week 7 to Week 10)

AC Circuit Analysis

- AC circuits with sinusoidal inputs
- Use of phasors and complex impedance in AC circuit analysis
- AC power: real, reactive and apparent power
- Power factor (leading/lagging)
- Series and parallel resonance
- Transformers and coupled coils
- Application of network theorems: Two port networks

ELEC2134 : Circuits and Signals (PART A)

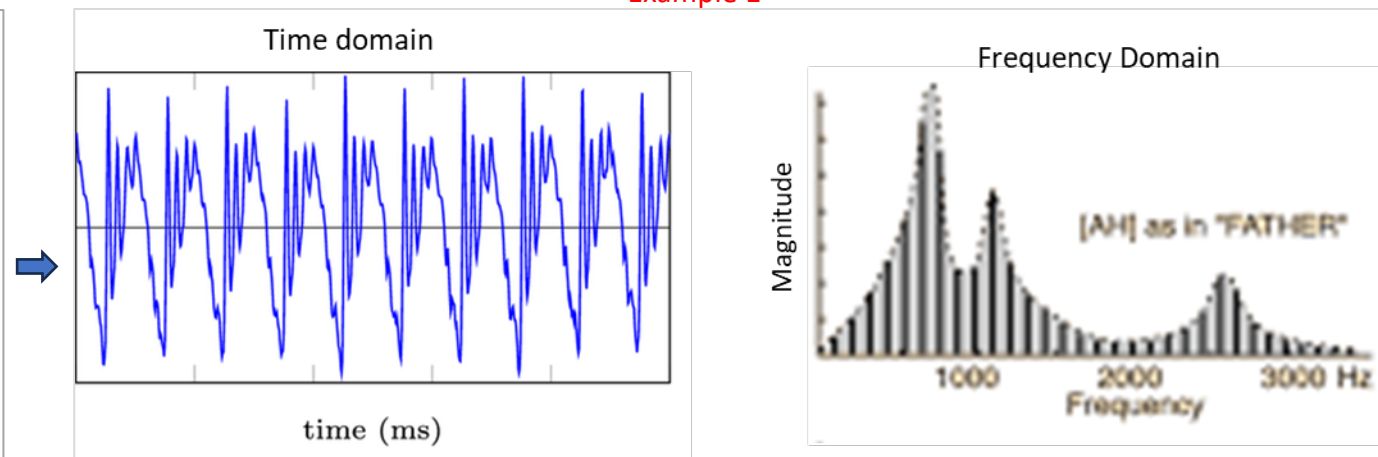
Topic 2: Transform Methods: Aperiodic signals and Fourier transforms

The **Fourier Transform** shows what frequencies are present in a signal and how much each frequency contributes.

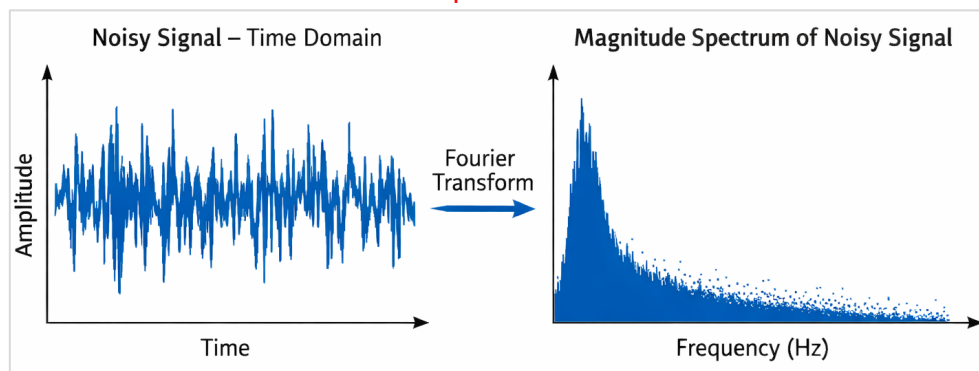
Time Domain vs Frequency Domain

- In the time domain, we see the speech waveform, but we don't know what frequencies it contains.
- The Fourier Transform converts this waveform into the frequency domain, where we clearly see the pitch and formants (three peaks) that define the vowel.

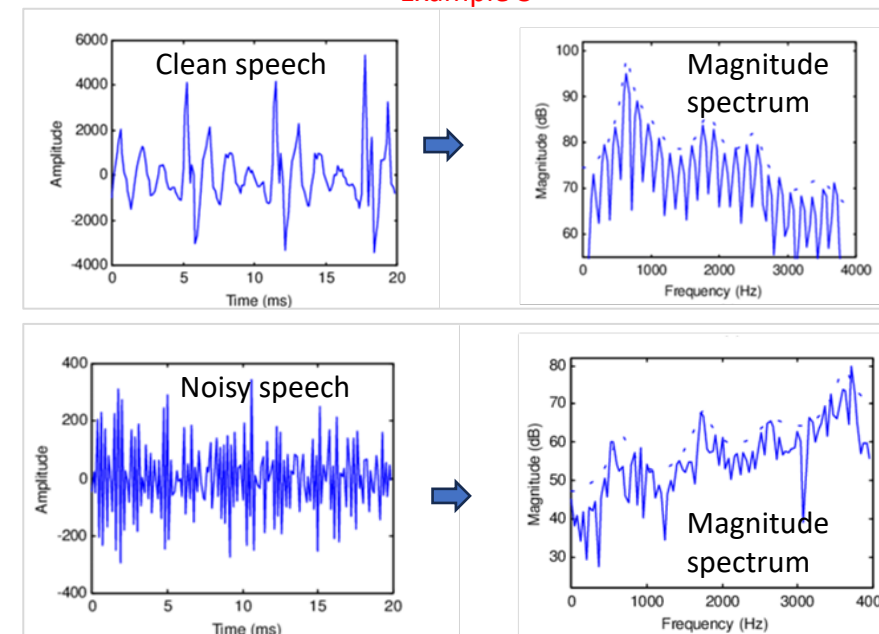
Example 1



Example 2



Example 3

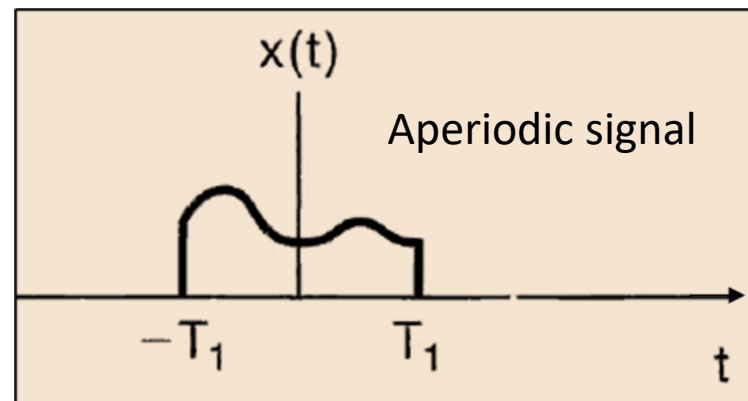
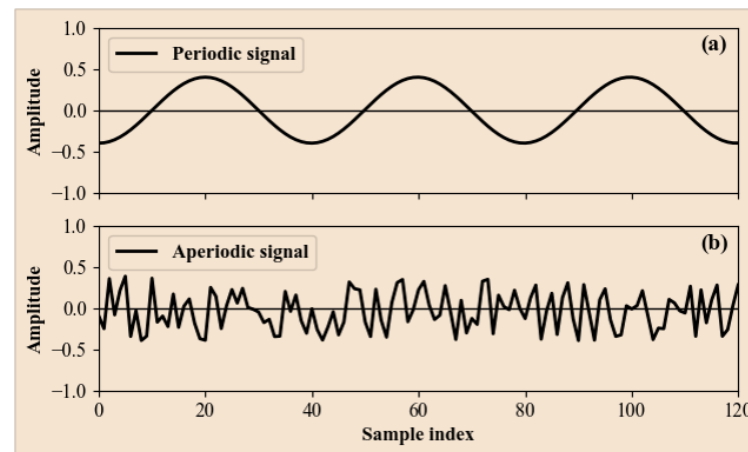


Periodic and Aperiodic Signals

- A **periodic signal** repeats the same pattern again and again:
$$x(t) = x(t + nT), n = \pm 1, \pm 2, \pm 3, \dots$$
 The **period** is T .

The frequency components of a periodic signal are analysed using a **Fourier series**.

- An aperiodic signal has no repeating pattern, and we assume its period is infinite ($T \rightarrow \infty$). Most signals of practical importance are aperiodic.
- Recall that a periodic waveform $x(t)$ possesses a Fourier series. As we increase the period T , the fundamental frequency ω_0 becomes smaller since, $\omega_0 = \frac{2\pi}{T}$
- As T is increased indefinitely, the individual spectral components merge into a continuous spectrum and the fundamental frequency becomes vanishingly small.
- Consequently, the frequency $n\omega_0$ of each harmonic component becomes the continuous frequency variable ω , the line spacing ω_0 becomes the infinitesimal $d\omega$ and the operation of summation becomes the operation of integration.



Period is infinite ($T \rightarrow \infty$)

Periodic and Aperiodic Signals

- Consider the exponential Fourier Series

$$x(t) = \sum_{n=-\infty}^{\infty} C_n e^{jn\omega_0 t}; \quad C_n = \frac{1}{T} \int_{-T/2}^{T/2} x(t) e^{-jn\omega_0 t} dt \quad \text{where } \omega_0 = \frac{2\pi}{T}$$

$$x(t) = \sum_{n=-\infty}^{\infty} \left\{ \frac{1}{T} \int_{-T/2}^{T/2} x(t) e^{-jn\omega_0 t} dt \right\} e^{jn\omega_0 t}; \quad f(t) = \sum_{n=-\infty}^{\infty} \left\{ \frac{1}{2\pi} \int_{-T/2}^{T/2} x(t) e^{-jn\omega_0 t} dt \right\} e^{jn\omega_0 t} \omega_0;$$

$$\left\{ T \rightarrow \infty; \quad n\omega_0 \rightarrow \omega; \quad \omega_0 \rightarrow d\omega; \quad \sum \rightarrow \int \right\}$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \left\{ \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt \right\} e^{j\omega t} d\omega$$

- The integral within brackets is called the Fourier Transform (FT) of $x(t)$ and is denoted by $X(\omega)$ or $X(j\omega)$;

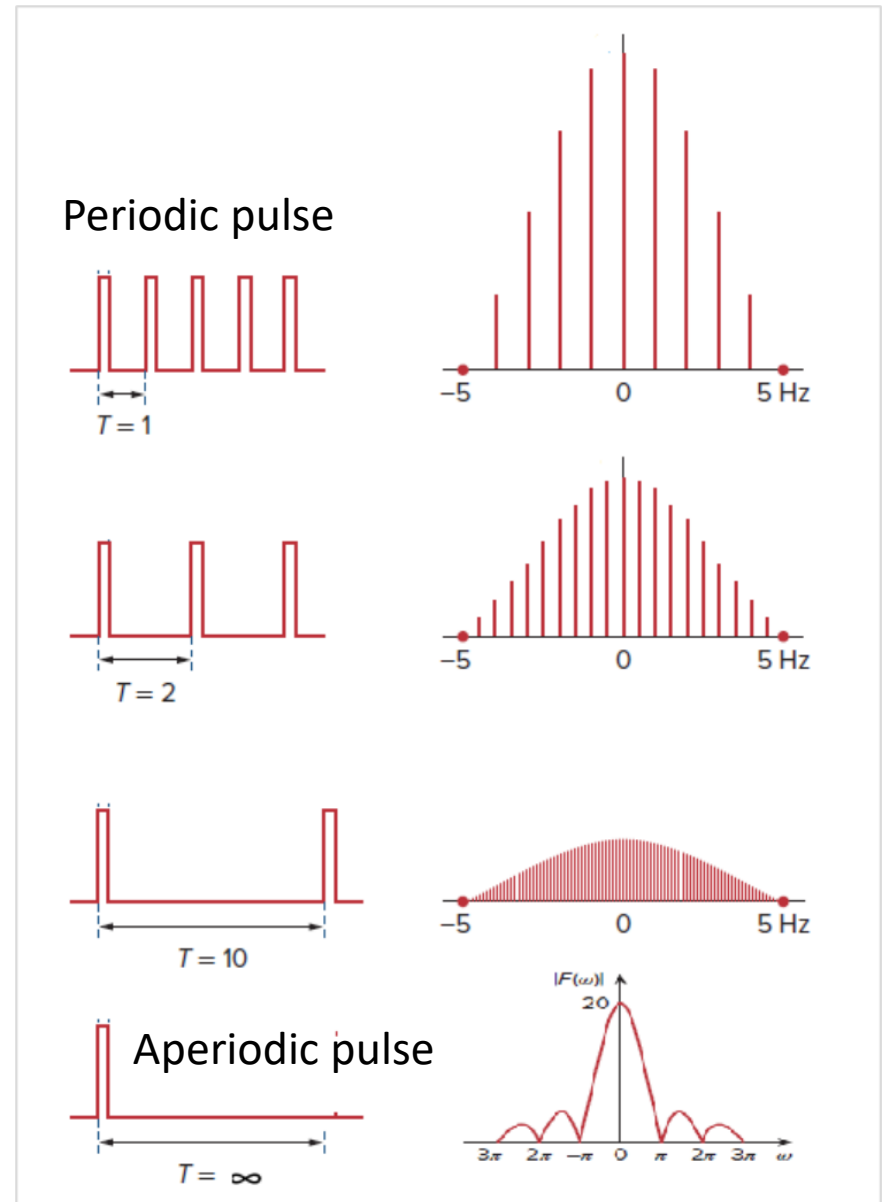
$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt \quad (\text{FT}) \quad (X(\omega) \text{ is a complex valued function})$$

- The Inverse Fourier Transform (IFT) of $X(\omega)$ is given by:

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega \quad (\text{IFT}) \quad (x(t) \text{ is normally a real valued signal})$$

From Fourier Series to Fourier Transform

- The figure illustrates the transition from a Fourier series to the Fourier transform as the period T of a periodic pulse train increases.
- For small T , the signal repeats frequently in time and its frequency spectrum consists of discrete spectral lines that are widely spaced. As T increases, the pulses move further apart in time, the fundamental frequency decreases, and the spacing between spectral lines becomes smaller.
- Consequently, more frequency components appear within a given frequency range, and the discrete spectrum begins to resemble a continuous one.
- In the limit as $T \rightarrow \infty$, the periodic signal becomes an aperiodic single pulse, the spectral lines merge into a continuous spectrum, and the Fourier series representation naturally transitions to the Fourier transform.



Fourier Transform (FT)

- The Fourier Transform pair is given by:

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt \quad (\text{FT})$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega \quad (\text{IFT})$$

Model answer

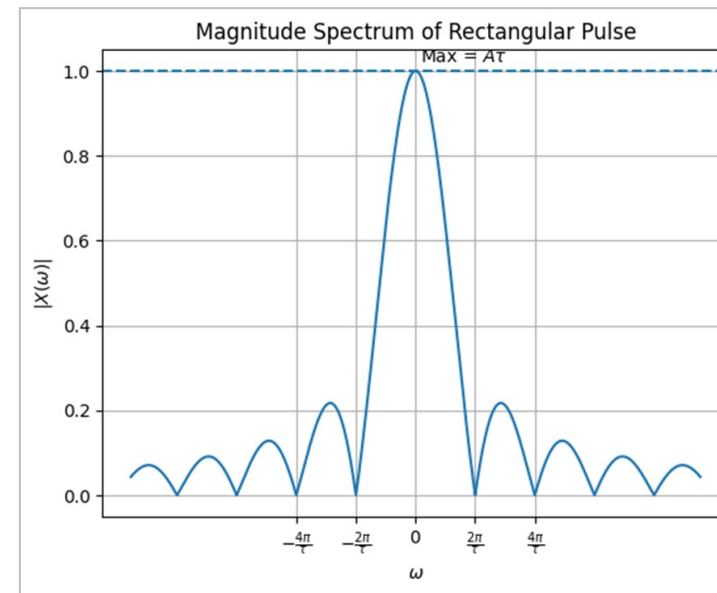
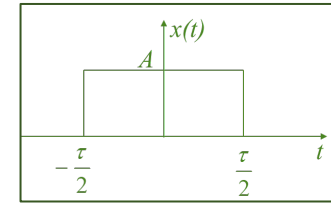
$$X(\omega) = \int_{-\tau/2}^{\tau/2} A e^{-j\omega t} dt = -\frac{A}{-j\omega} [e^{-j\omega\tau/2} - e^{j\omega\tau/2}]$$

$$X(\omega) = A\tau \frac{\sin(\frac{\omega\tau}{2})}{\frac{\omega\tau}{2}} \quad (\text{Sinc function})$$

The magnitude spectrum $|X(\omega)|$ is sinc-shaped, with maximum value $A\tau$ at $\omega = 0$, and nulls at integer multiples of $\frac{2\pi}{\tau}$.

Example 1

- Evaluate the Fourier transform of a rectangular pulse shown below:



A rectangular time-domain pulse is transformed to a 'sinc' function in frequency domain.

The Fourier transform is a continuous function of frequency and $|X(\omega)|$ represents the distribution of amplitude with frequency

Fourier Transform (FT)

Example 2

- Find the FT of $x(t) = e^{-at}u(t)$ $a > 0$

Model answer

$$X(\omega) = \int_{-\infty}^{\infty} e^{-at} u(t) e^{-j\omega t} dt = \frac{1}{a+j\omega} \quad (\text{Note: } X(\omega) \text{ is complex})$$

Magnitude: $|X(\omega)| = \frac{1}{\sqrt{a^2 + \omega^2}};$

Phase: $\arg(X(\omega)) = -\tan^{-1}\left(\frac{\omega}{a}\right)$

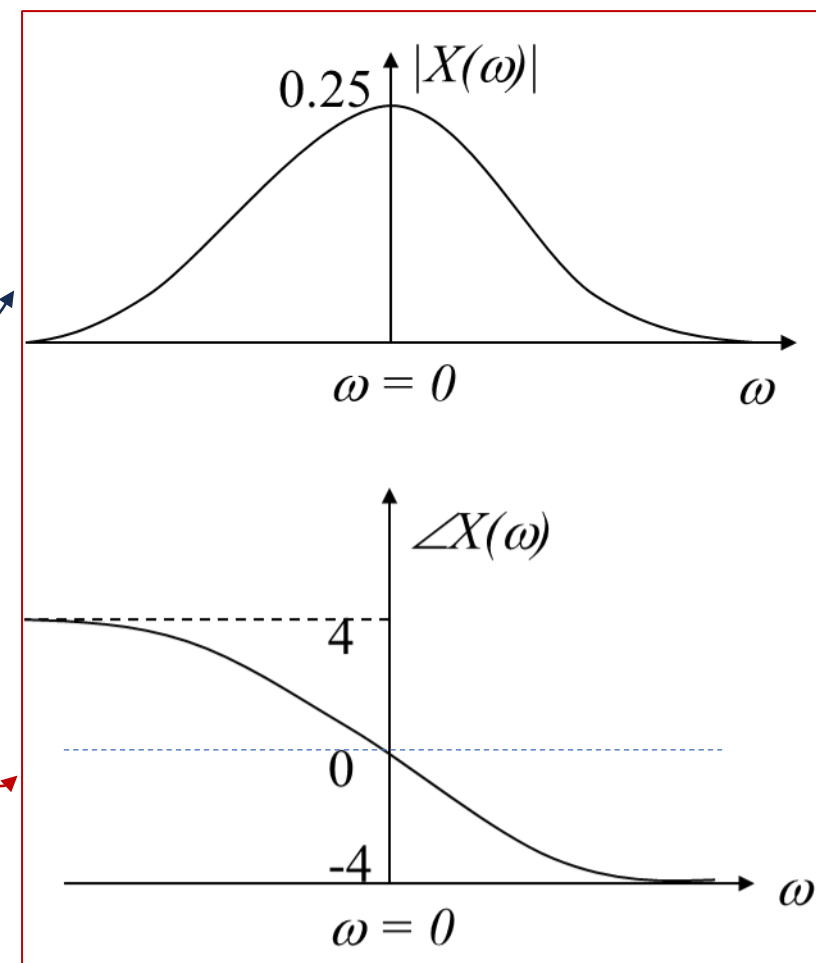
Example 3

- Find the FT of $x(t) = te^{-2t}u(t)$

Model answer

$$\begin{aligned} X(\omega) &= \int_0^{\infty} te^{-2t} e^{-j\omega t} dt \\ &= \int_0^{\infty} te^{-(2+j\omega)t} dt \\ &= \frac{1}{(2+j\omega)^2} \end{aligned}$$

$$\begin{aligned} |X(\omega)| &= \frac{1}{\sqrt{(4-\omega^2)^2 + (4\omega)^2}} \\ \angle X(\omega) &= -\tan^{-1}\left(\frac{4\omega}{4-\omega^2}\right) \end{aligned}$$

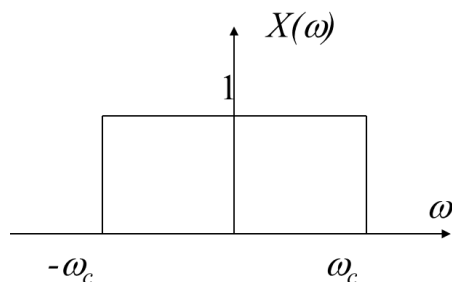


Inverse Fourier Transform (IFT)

Example 4

- Find the **inverse** FT of the rectangular spectrum shown below:

$$X(\omega) = \begin{cases} 1 & -\omega_c \leq \omega \leq \omega_c \\ 0 & |\omega| > \omega_c \end{cases}$$



Fourier Transform pair :

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt \quad (\text{FT})$$

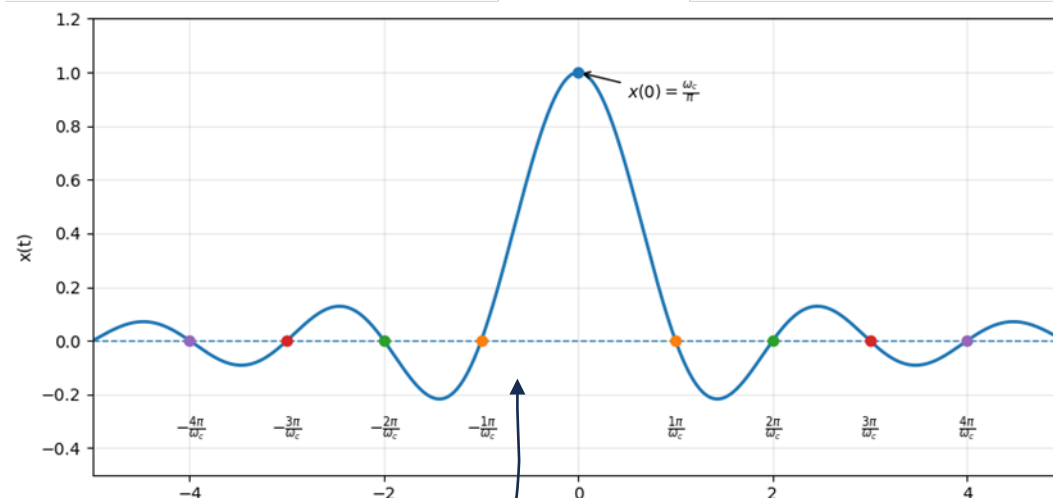
$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega \quad (\text{IFT})$$

Model answer

$$x(t) = \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} X(\omega) e^{j\omega t} d\omega; \quad x(t) = \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} 1 \cdot e^{j\omega t} d\omega$$

$$x(t) = \frac{1}{\pi t} \sin(\omega_c t) = \frac{\omega_c}{\pi} \frac{\sin \omega_c t}{\omega_c t}$$

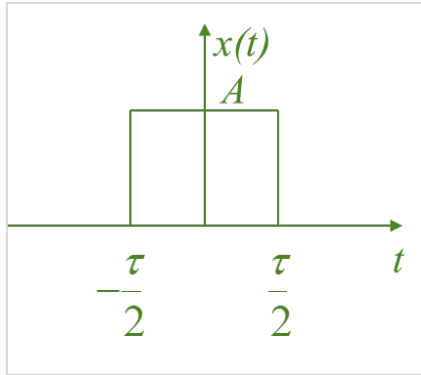
$$\text{At } t = 0 \quad \frac{\sin \omega_c t}{\omega_c t} = 1 \quad \text{and} \quad x(0) = \frac{\omega_c}{\pi}$$



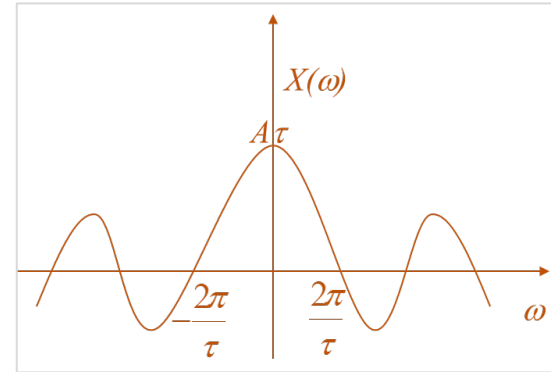
A 'sinc' function in time is transformed to a rectangular function in frequency domain

Short Summary

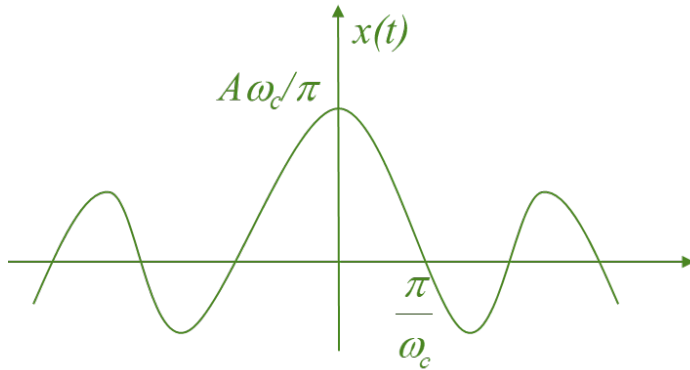
- A rectangular pulse in the time domain transforms into a Sinc function in the frequency domain.



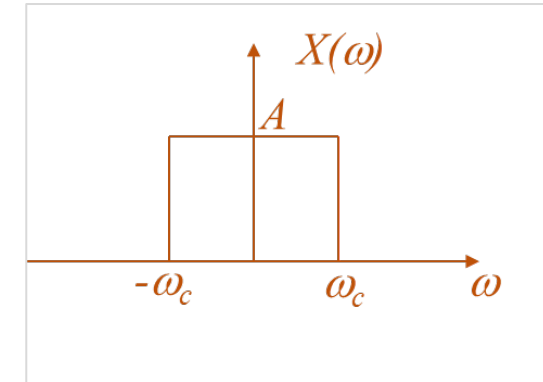
FT
IFT



- The Fourier transform of a sinc function in time is a rectangular function in frequency.



FT
IFT

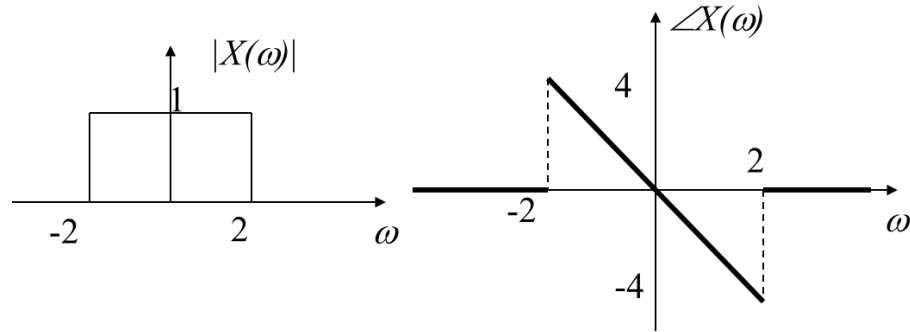


Rectangle in time $\Leftrightarrow$ sinc in frequency
sinc in time $\Leftrightarrow$ rectangle in frequency

Inverse Fourier Transform (IFT)

Example 5

- Determine the time-domain signals corresponding to the following Fourier Transform $X(\omega)$:



Fourier Transform pair :

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt \quad (\text{FT})$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega \quad (\text{IFT})$$

Model answer

- The phase line passes through (0,0) and has a slope (m)

$$m = \frac{-4-4}{2-(-2)} = -2, \therefore \angle X(\omega) = -2\omega \text{ for } |\omega| \leq 2.$$

- If the rectangle height is 1 (as typically drawn), then

$$X(\omega) = \begin{cases} e^{-j2\omega}, & |\omega| \leq 2 \\ 0, & |\omega| > 2 \end{cases}$$

- Using inverse Fourier Transform $x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega$

- We get

$$x(t) = \frac{1}{2\pi} \int_{-2}^2 e^{-j2\omega} e^{j\omega t} d\omega$$

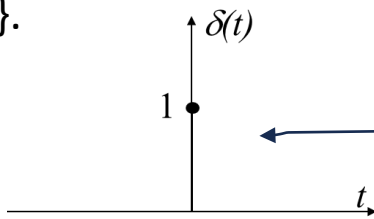
$$= \frac{1}{2\pi} \int_{-2}^2 e^{j\omega(t-2)} d\omega$$

$$= \frac{1}{\pi} \frac{\sin(2(t-2))}{\pi(t-2)}$$

Fourier Transform examples

Example 6

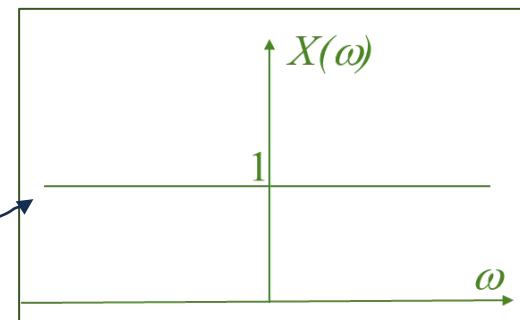
- Find the Fourier Transform of $x(t) = \delta(t)$ {Delta function or Impulse}.



Model answer

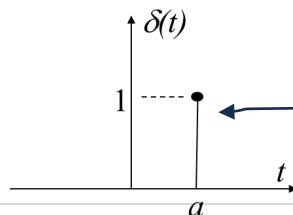
$$X(\omega) = \int_{-\infty}^{\infty} 1 e^{-j\omega t} dt = 1$$

A delta function in time contains all frequencies with equal amplitude



Example 7

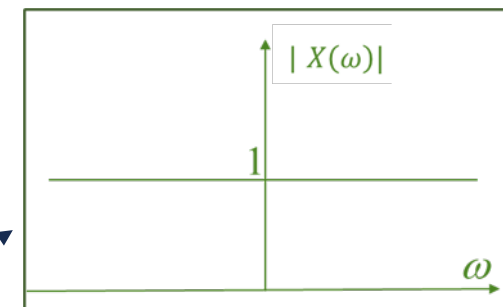
- Find the Fourier Transform of $x(t) = \delta(t-a)$ {Delta function or Impulse}.



Model answer

$$X(\omega) = \int_{-\infty}^{\infty} 1 e^{-j\omega t} dt = e^{-j\omega a}$$

A shifted delta function in time contains all frequencies with equal amplitude



Note

$$FT\{\delta(t-a)\} = e^{-j\omega a}; \quad \therefore IFT\{e^{-j\omega a}\} = \delta(t-a)$$

$$\text{Proof: } IFT\{e^{-j\omega a}\} = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-j\omega a} e^{j\omega t} d\omega = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{j\omega(t-a)} d\omega = \delta(t-a)$$

$$\therefore \int_{-\infty}^{\infty} e^{j\omega(t-a)} d\omega = 2\pi\delta(t-a); \text{ if } a = 0 \Rightarrow \int_{-\infty}^{\infty} e^{j\omega t} d\omega = 2\pi\delta(t)$$

$$\text{By interchanging variables, we have } \int_{-\infty}^{\infty} e^{j\omega t} dt = 2\pi\delta(\omega)$$

Fourier Transform pair:

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt \quad (\text{FT})$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega \quad (\text{IFT})$$

Fourier Transform Examples

Example 8

- Find the Fourier Transform of $x(t) = e^{j\omega_1 t}$

Model answer

$$\text{FT}\{e^{j\omega_1 t}\} = X(\omega) = \int_{-\infty}^{\infty} e^{j\omega_1 t} e^{-j\omega t} dt$$

$$X(\omega) = \int_{-\infty}^{\infty} e^{j(\omega_1 - \omega)t} dt = 2\pi\delta(\omega_1 - \omega)$$

$$\text{FT}\{e^{j\omega_1 t}\} = 2\pi\delta(\omega - \omega_1)$$

Note

- Prove that $\delta(\omega - \omega_1) = \delta(\omega_1 - \omega)$

We know that $\delta(x) = \delta(-x)$ (even function)

Now note that:

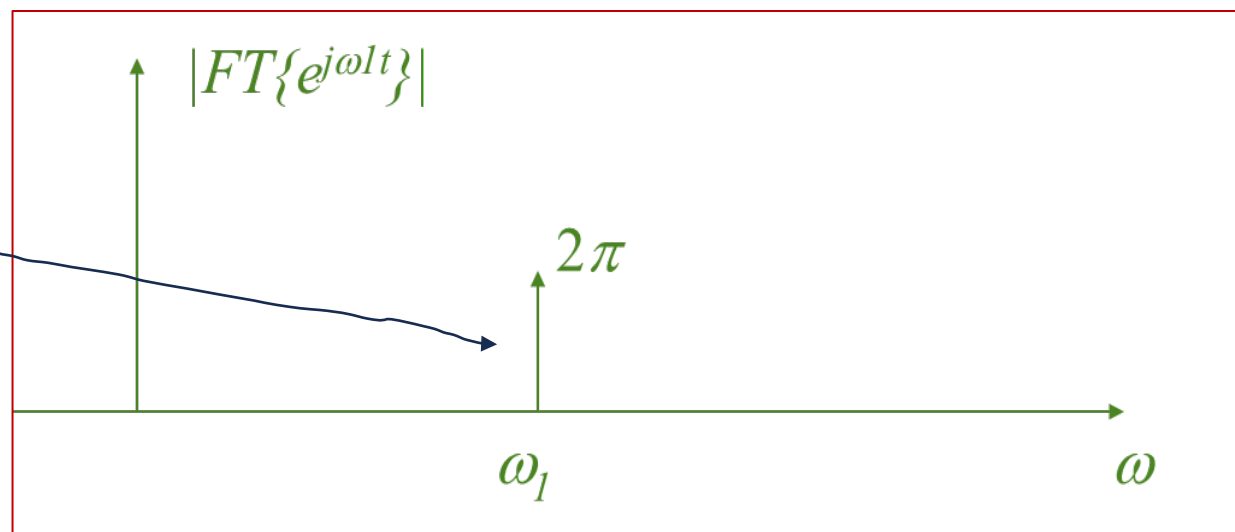
$$\omega_1 - \omega = -(\omega - \omega_1)$$

So

$$\delta(\omega_1 - \omega) = \delta(-(\omega - \omega_1)) = \delta(\omega - \omega_1)$$

$$\int_{-\infty}^{\infty} e^{j\omega t} dt = 2\pi\delta(\omega)$$

The magnitude spectrum of Fourier Transform of $e^{j\omega_1 t}$ is shown below



Fourier Transform Examples

Example 9

- Find the Fourier Transform of $x(t) = \cos(\omega_1 t)$

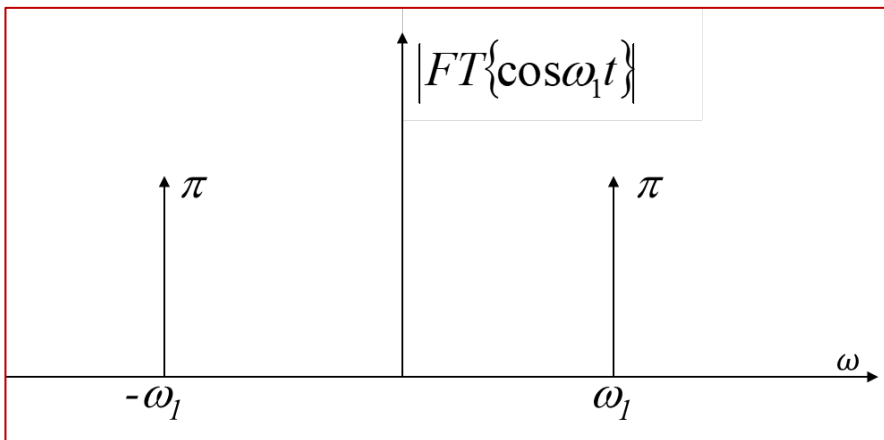
Model answer

$$FT\{\cos(\omega_1 t)\} = X(\omega) = FT\left\{\frac{e^{j\omega_1 t} + e^{-j\omega_1 t}}{2}\right\}$$

$$X(\omega) = \frac{1}{2}FT\{e^{j\omega_1 t}\} + \frac{1}{2}FT\{e^{-j\omega_1 t}\}$$

$$FT\{\cos(\omega_1 t)\} = \pi\delta(\omega - \omega_1) + \pi\delta(\omega + \omega_1)$$

The magnitude spectrum of Fourier Transform of $\cos(\omega_1 t)$ is shown below



Summary

Fourier Transform pair:

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt \quad (\text{FT})$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega \quad (\text{IFT})$$

$$FT\{\delta(t)\} = 1;$$

$$FT\{A \delta(t)\} = A$$

$$FT\{\delta(t-a)\} = e^{-j\omega a}$$

$$FT\{\delta(t+a)\} = e^{j\omega a}$$

$$\int_{-\infty}^{\infty} e^{j\omega t} d\omega = 2\pi\delta(t); \quad \int_{-\infty}^{\infty} e^{j\omega t} dt = 2\pi\delta(\omega)$$

$$FT\{e^{j\omega_1 t}\} = 2\pi\delta(\omega - \omega_1)$$

$$FT\{e^{-j\omega_1 t}\} = 2\pi\delta(\omega + \omega_1)$$

$$FT\{A\} = 2\pi A\delta(\omega)$$

$$FT\{\cos(\omega_1 t)\} = \pi\delta(\omega - \omega_1) + \pi\delta(\omega + \omega_1)$$

$$FT\{\sin(\omega_1 t)\} = j\pi[\delta(\omega + \omega_1) - \pi\delta(\omega - \omega_1)]$$

Time and frequency domain analysis

Example 10: Multiplication of sinusoids

- The product of two real sinusoids (frequencies f_1 kHz and f_2 kHz) results in the sum and difference frequencies $[(f_1 + f_2)$ kHz and $(f_1 - f_2)$ kHz]

- Consider a signal $x(t)$ which is a product of two sinusoids, as given by:

$$x(t) = \sin(\pi t) \cdot \sin(10\pi t)$$

$x(t)$ can be re-written as

$$x(t) = \left(\frac{e^{j\pi t} - e^{-j\pi t}}{2j} \right) \left(\frac{e^{j10\pi t} - e^{-j10\pi t}}{2j} \right) = \frac{-1}{4} (e^{j\pi t} - e^{-j\pi t})(e^{j10\pi t} - e^{-j10\pi t})$$

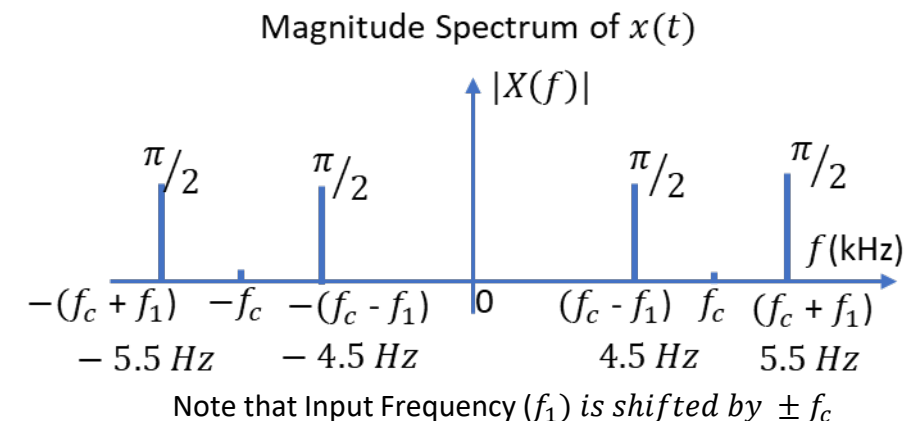
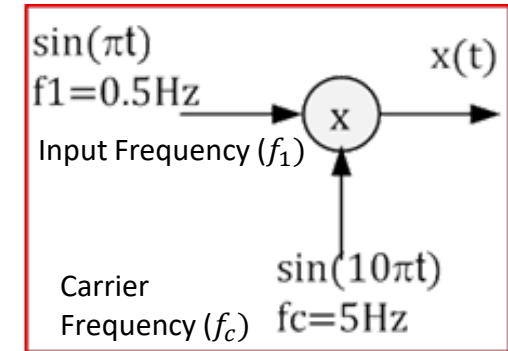
$$x(t) = \frac{-1}{4} [e^{j11\pi t} - e^{-j9\pi t} - e^{j9\pi t} + e^{-j11\pi t}] = \frac{1}{2} \cos(9\pi t) - \frac{1}{2} \cos(11\pi t) = \frac{1}{2} \cos(2\pi 4.5t) - \frac{1}{2} \cos(2\pi 5.5t)$$

- Recall example 10: $FT\{\cos \omega_1 t\} = \pi\delta(\omega - \omega_1) + \pi\delta(\omega + \omega_1)$

From $x(t)$ it can be seen that there are four spectral

components at frequencies $f_c + f_1 = 5.5$ Hz, $f_c - f_1 = 4.5$ Hz,

$-(f_c + f_1) = -4.5$ Hz, $-(f_c - f_1) = -5.5$ Hz, as shown on the right.



ELEC2134 : Circuits and Signals (PART A)

Fourier Transform Properties

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Term 3, 2026

Fourier Transform Properties – Translation in the frequency domain

1. Frequency – Shifting Property

$$e^{j\omega_0 t} x(t) \xleftrightarrow{FT} X(\omega - \omega_0)$$

Example 11

- Using the frequency-shift property to determine the FT of the complex sinusoidal pulse.

$$y(t) = \begin{cases} e^{j10t} & |t| \leq \pi \\ 0 & \text{otherwise} \end{cases}$$

Model answer

- We express $y(t)$ as the product of a complex sinusoid e^{j10t} and a rectangular pulse

$$x(t) = \begin{cases} 1 & |t| \leq \pi \\ 0 & \text{otherwise} \end{cases}$$

We obtain,

$$x(t) \xleftrightarrow{FT} X(\omega) = 2 \frac{\sin(\omega\pi)}{\omega} \quad (\text{see Page 7})$$

Using the shifting property $e^{j10t} x(t) \xleftrightarrow{FT} X(\omega - 10)$

We obtain $y(t) \xleftrightarrow{FT} \frac{2}{\omega - 10} \sin(\omega - 10)\pi$

$$Y(\omega) = 2 \frac{\sin(\omega - 10)\pi}{(\omega - 10)}$$

Fourier Transform Properties – Translation in the Time Domain

2. Time – Shifting Property

$$x(t - t_0) \xleftrightarrow{FT} e^{-j\omega t_0} X(\omega)$$

Example 12

- Using the Fourier transform (FT) of the rectangular pulse $x(t)$ shown in Figure 1(a), determine the FT of the time-shifted rectangular pulse $y(t)$ shown in Figure 1(b).

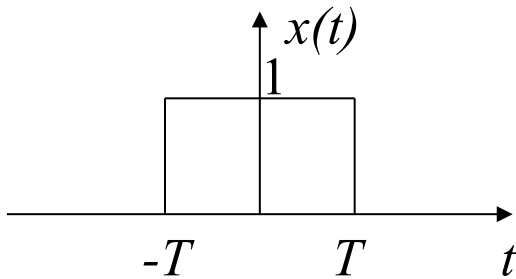


Fig 1(a)

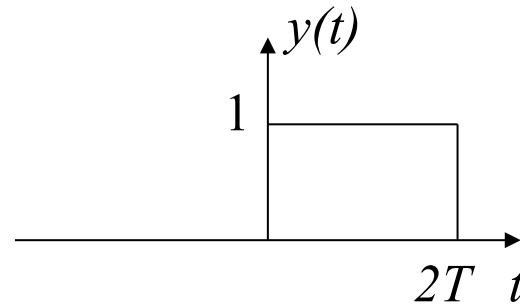


Fig 1(b)

Model answer

- Note that $y(t) = x(t - T)$. By the time-shift property of the Fourier transform

$$Y(\omega) = e^{-j\omega T} X(\omega)$$

Therefore, $X(\omega) = 2 \frac{\sin \omega T}{\omega}$, $\tau = T$

Thus we have, $Y(\omega) = e^{-j\omega T} \times \frac{2}{\omega} \sin \omega T$

Fourier Transform Properties – Scale change

3. Scaling Property

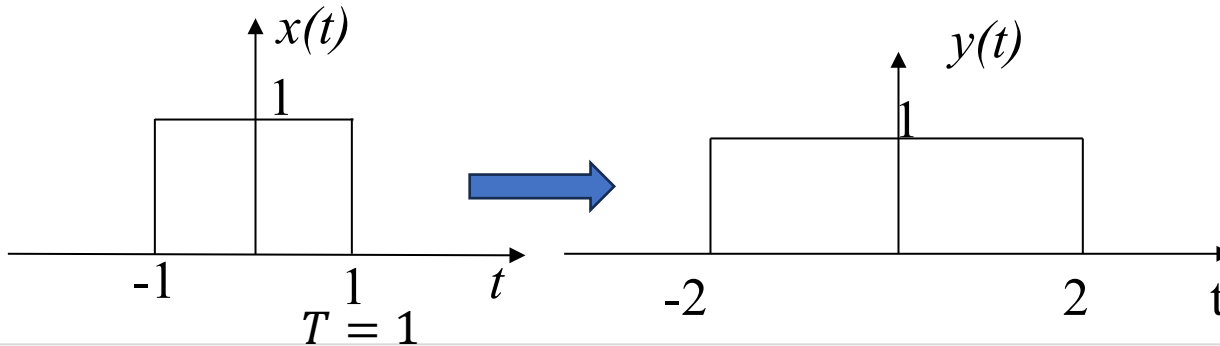
$$\text{If } y(t) = x(at), \text{ then } Y(\omega) = \frac{1}{|a|} X\left(\frac{\omega}{a}\right)$$

Example 13

- Let $x(t)$ be the rectangular pulse $x(t) = \begin{cases} 1 & |t| \leq 1 \\ 0 & |t| > 1 \end{cases}$

Use the Fourier transform of $x(t)$ and the scaling property to find the Fourier transform of the scaled rectangular pulse

$$y(t) = \begin{cases} 1 & |t| \leq 2 \\ 0 & |t| > 2 \end{cases}$$



Model answer

$$X(\omega) = 2 \frac{\sin(\omega T)}{\omega} \quad (\text{Page 7 for rectangular pulse})$$

$$T = 1, \quad \therefore X(\omega) = 2 \frac{\sin(\omega)}{\omega}$$

Note that $y(t) = x\left(\frac{1}{2}t\right)$, Applying the scaling property of the Fourier transform gives (Note $a = \frac{1}{2}$)

$$y(t) = x\left(\frac{t}{2}\right) \Rightarrow Y(\omega) = \frac{1}{|a|} X\left(\frac{\omega}{a}\right) = 2 X(2\omega)$$

$$= 2 \cdot \frac{2}{2\omega} \sin(2\omega)$$

$$= \frac{2}{\omega} \sin(2\omega)$$

Fourier Transform Properties – Differentiation in Time

4. Differentiation in Time

$$\frac{d}{dt} x(t) \xleftrightarrow{FT} j\omega X(\omega)$$

$$\text{The } n\text{th derivative of } x(t): \frac{d^n}{dt^n} x(t) \xleftrightarrow{FT} (j\omega)^n X(\omega)$$

Example 14

$$\text{Let } x(t) = e^{-at}u(t) \Rightarrow X(\omega) = \frac{1}{j\omega + a}$$

$$\therefore \frac{d}{dt} (e^{-at}u(t)) \xleftrightarrow{FT} j\omega X(\omega); \quad X(\omega) = j\omega \frac{1}{j\omega + a} = \frac{j\omega}{j\omega + a}$$

Let's verify this result by differentiating and finding the Fourier transform directly:

$$\begin{aligned} \frac{d}{dt} (e^{-at}u(t)) &= -ae^{-at}u(t) + e^{-at}\delta(t) \\ &= -ae^{-at}u(t) + \delta(t) \end{aligned}$$

Taking the Fourier transform, we obtain

$$\text{FT} \{-ae^{-at}u(t) + \delta(t)\} = \frac{-a}{j\omega + a} + 1 = \frac{j\omega}{a + j\omega}$$

Fourier Transform Properties

5. Differentiation in Frequency

$$-jtx(t) \xleftrightarrow{FT} \frac{d}{d\omega} X(\omega)$$

Differentiation in Frequency corresponds to multiplication in time by jt

$$\text{Alternatively: } tx(t) \xleftrightarrow{FT} j \frac{d}{d\omega} X(\omega)$$

[Known as time multiplication]

$$\text{Alternatively: } t^n x(t) \xleftrightarrow{FT} j^n \frac{d^n}{d\omega^n} X(\omega)$$

Example 15

If $x(t) = e^{-a|t|}$ $a > 0$, Find the FT of $\{-jtx(t)\}$

$$X(\omega) = \int_{-\infty}^{\infty} e^{-a|t|} e^{-j\omega t} dt = \frac{2a}{a^2 + \omega^2}$$

$$\text{Applying the FT property } -jtx(t) \xleftrightarrow{FT} \frac{d}{d\omega} X(\omega)$$

$$\text{Differentiate } X(\omega): \frac{d}{d\omega} X(\omega) = \frac{d}{d\omega} \left(\frac{2a}{a^2 + \omega^2} \right) = \frac{-4a\omega}{(a^2 + \omega^2)^2}$$

$$\text{Final result: } -jt e^{-a|t|} \xleftrightarrow{FT} \frac{-4a\omega}{(a^2 + \omega^2)^2}$$

6. Linearity

$$\text{Time domain: } y(t) = a x_1(t) + b x_2(t)$$

$$\text{Frequency Domain: } Y(\omega) = a X_1(\omega) + b X_2(\omega)$$

Scaling and adding signals in the time domain results in the same scaling and addition of their Fourier transforms in the frequency domain.

7. Time Integration

$$\int_{-\infty}^t x(\tau) d\tau \xleftrightarrow{FT} \frac{X(\omega)}{j\omega}$$

This property holds if $\int_{-\infty}^{\infty} x(\tau) d\tau = 0$

Time integration of a function $x(t)$ corresponds to division of its Fourier Transform $X(\omega)$ by $j\omega$

8. Reversal

$$x(-t) \xleftrightarrow{FT} X(-\omega) = X^*(\omega)$$

Reversing $x(t)$ about the time axis reverses $X(\omega)$ about the frequency axis. (* stands for Complex value).

Fourier Transform Properties – Convolution in the Time Domain

Convolution describes how the output of a system depends on the **input signal and the system's impulse response**. If:

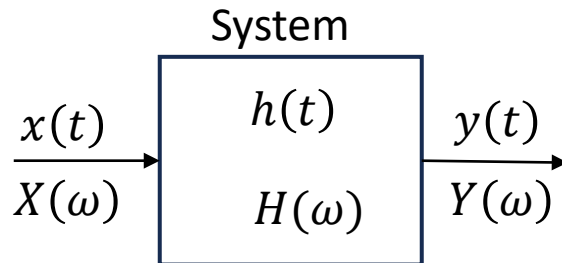
$x(t)$ = input signal

$h(t)$ = impulse response of a system

$y(t)$ = output signal

then the output is given by the convolution integral:

$$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau) h(t - \tau) d\tau$$



$$y(t) = x(t) * h(t)$$

Thus, the **convolution** in the **time domain** corresponds to **multiplication** in the **frequency domain**.

9. Convolution in the time domain

$$\int_{-\infty}^{\infty} x(\tau) h(t - \tau) d\tau \xleftrightarrow{FT} X(\omega) H(\omega)$$

Convolution in the Time Domain corresponds to multiplication in the frequency domain.

$$x(t) * h(t) \leftrightarrow X(\omega) \cdot H(\omega)$$

** convolution*

Note: When the input is an impulse, $x(t) = \delta(t)$, since $X(\omega) = 1$, we obtain the impulse response:

$$Y(\omega) = H(\omega)$$

Taking IFT, we obtain $y(t) = h(t)$.

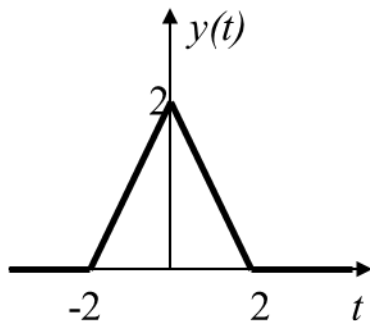
Fourier Transform Properties

Example 16

Given the Fourier transform pair

$$x(t) = \begin{cases} 1, & |t| \leq 1 \\ 0, & \text{otherwise} \end{cases} \Leftrightarrow X(\omega) = \frac{2 \sin \omega}{\omega},$$

use the **convolution property** to find the frequency-domain representation $Y(\omega)$ for the following signal $y(t)$.



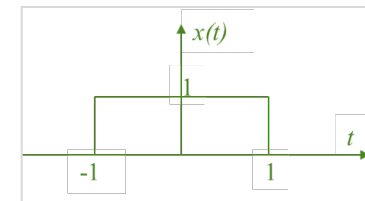
Model Answer

$$x(t) * x(t) \xrightarrow{FT} X(\omega) \cdot X(\omega)$$

$$\begin{aligned} \therefore Y(\omega) &= X(\omega) \cdot X(\omega) \\ &= \frac{4}{\omega^2} \sin^2 \omega \end{aligned}$$

Note: Why does the convolution of two rectangles give a triangle?

✓ $y(t) = x(t) * x(t) = \int_{-\infty}^{\infty} x(\tau) x(t - \tau) d\tau$



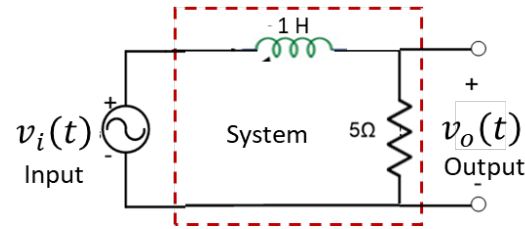
- ✓ Convolution measures **how much overlap** there is between one rectangle and a shifted version of the other.
- ✓ Imagine: One rectangle is **fixed** ($x(\tau)$) and the other rectangle slides across it ($x(t - \tau)$)
- ✓ The convolution output $y(t)$ at time t is the **area of overlap** between the two rectangles.
- ✓ No overlap occurs when the rectangles are far apart: $y(t) = 0$
- ✓ As the second rectangle starts overlapping: Overlap length increases linearly, so $y(t)$ increases linearly and this creates the rising edge of the triangle.
- ✓ When perfectly aligned: Overlap length = full width = 2, So $y(0) = 2$
This is the peak of the triangle.

Fourier Transform Properties

Example 17

- Find the response $v_o(t)$ of the RL circuit shown below when the input voltage $v_i(t) = 4e^{-2t}u(t)$ Volts

The initial condition is zero.



Model Answer

Since $v_i(t) = 4e^{-2t}u(t)$, we obtain

$$V(\omega) = \frac{4}{2 + j\omega} \quad (\text{FT of } v_i(t))$$

The circuit (System) is represented by $H(\omega)$, and using the voltage divider principle, we have

$$H(\omega) = \frac{R}{R + j\omega L} = \frac{5}{5 + j\omega}$$

Then,

$$V_o(\omega) = H(\omega) V(\omega) = \frac{5}{(5+j\omega)} \cdot \frac{4}{(2+j\omega)} = \frac{-20/3}{(5+j\omega)} + \frac{20/3}{(2+j\omega)}$$

Using the inverse transform for each term we have $v_o(t) = \frac{20}{3} (e^{-2t} - e^{-5t}) u(t)$ Volts

The time-domain responses obtained in this manner are responses of initially relaxed circuits.
(No initial energy is stored.)

Fourier Transform Properties - Modulation

10. Modulation

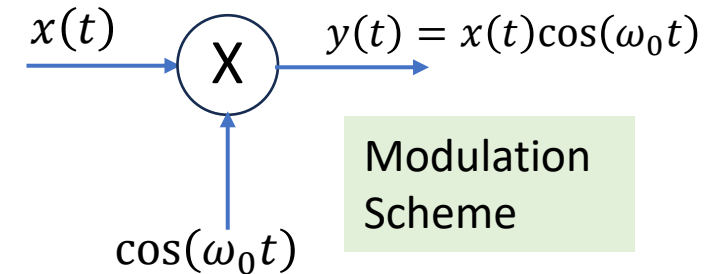
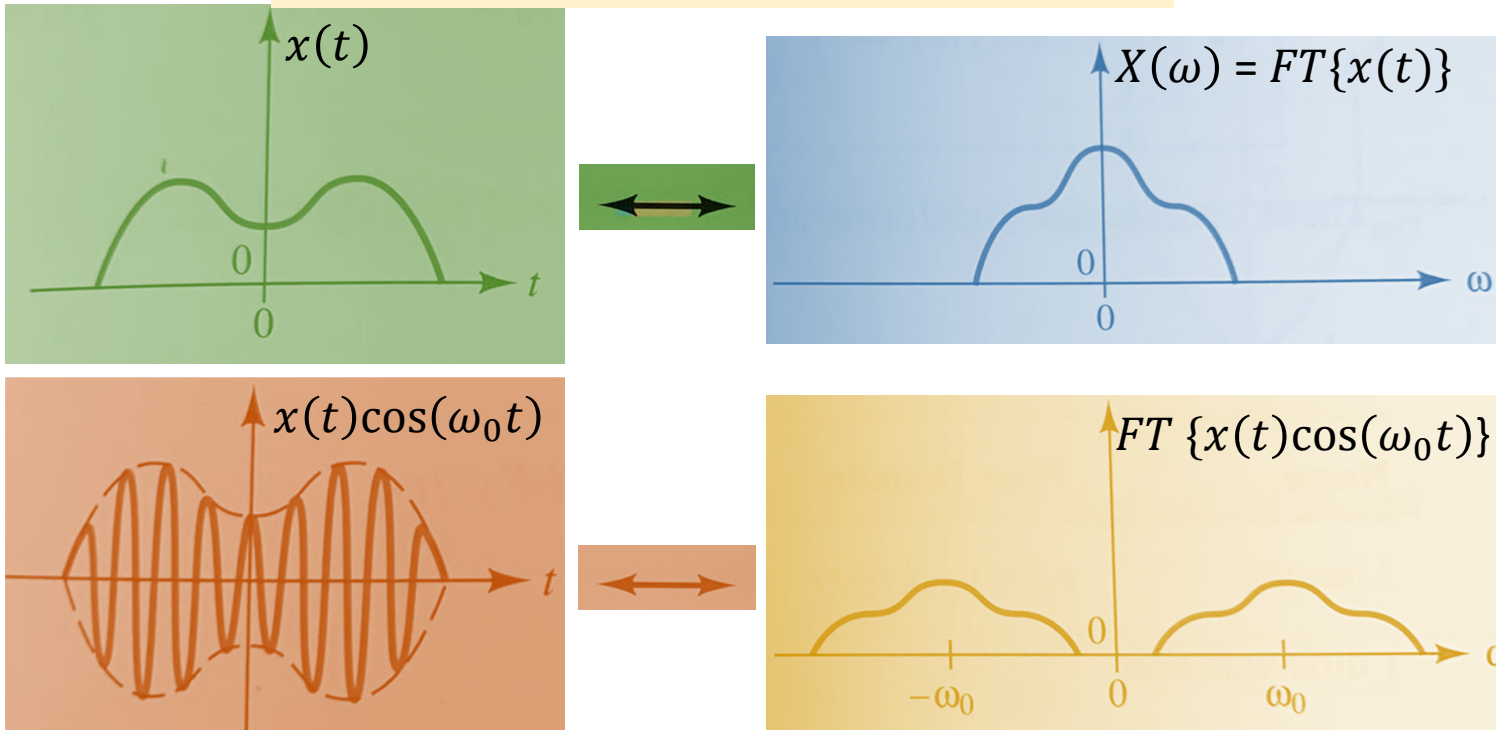
Amplitude modulation is the process of varying the amplitude of a sinusoidal carrier. If the modulating signal is denoted $x(t)$, the modulated carrier becomes

$$y(t) = x(t)\cos(\omega_0 t)$$

The amplitude spectrum of this carrier is one-half the amplitude spectrum of $x(t)$ shifted to be centred at $\pm\omega_0$, that is,

$$FT\{x(t)\cos(\omega_0 t)\} = \frac{1}{2} X(\omega - \omega_0) + \frac{1}{2} X(\omega + \omega_0)$$

The effect of modulation is shown below:



When a signal $x(t)$ with spectrum $X(\omega)$ modulates a sinusoidal carrier having frequency ω_0 , the resulting spectrum is the half the spectrum of $x(t)$ but shifted by $\pm\omega_0$

ELEC2134: Circuits and Signals

Appendix: Summary of Common Fourier Transform Pairs and Properties

Fourier Transform: Common Signals and Properties

Common Signals and Their Fourier Transforms

Type	$f(t)$	$F(\omega)$
Impulse	$\delta(t)$	1
Constant	A	$2\pi A\delta(\omega)$
Signum	$\text{sgn}(t)$	$\frac{2}{j\omega}$
Step	$u(t)$	$\pi\delta(\omega) + \frac{1}{j\omega}$
Positive-time exponential	$e^{-at}u(t), a > 0$	$\frac{1}{a + j\omega}, a > 0$
Negative-time exponential	$e^{at}u(-t), a > 0$	$\frac{1}{a - j\omega}, a > 0$
Positive & negative exponential	$(e^{-a t })$	$\frac{2a}{a^2 + \omega^2}$
Complex exponential	$e^{j\omega_0 t}$	$2\pi\delta(\omega - \omega_0)$
Cosine	$\cos \omega_0 t$	$\pi[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$
Sine	$\sin \omega_0 t$	$j\pi[\delta(\omega + \omega_0) - \delta(\omega - \omega_0)]$

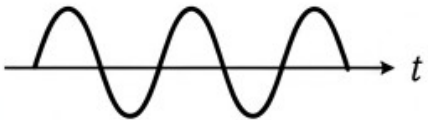
Fourier Transform Properties

Name of Property	Function of Time	Fourier Transform
1. Definition	$f(t)$	$F(\omega)$
2. Multiplication by constant	$Af(t)$	$AF(\omega)$
3. Linearity	$af_1(t) + bf_2(t)$	$aF_1(\omega) + bF_2(\omega)$
4. Time shift	$f(t - t_0)$	$e^{-j\omega t_0}F(\omega)$
5. Time scaling ($a > 0$)	$f(at)$	$\frac{1}{a}F\left(\frac{\omega}{a}\right)$
6. Modulation (frequency shift)	$e^{j\omega_0 t}f(t)$	$F(\omega - \omega_0)$
7. Differentiation (time)	$\frac{d^n f(t)}{dt^n}$	$(j\omega)^n F(\omega)$
8. Convolution (time)	$\int_{-\infty}^{\infty} f_1(x)f_2(t - x)dx$	$F_1(\omega)F_2(\omega)$
9. Time multiplication	$t^n f(t)$	$(j)^n \frac{d^n F(\omega)}{d\omega^n}$
10. Time reversal	$f(-t)$	$F(-\omega)$
11. Integration	$\int_{-\infty}^t f(\tau) d\tau$	$\frac{F(\omega)}{j\omega} + \pi F(0)\delta(\omega)$

Relationship Among Key Signal Representations and Transforms

Continuous-Time (CT) Signals $x(t)$

Periodic CT signal $x(t)$



$$x(t + T_0) = x(t)$$

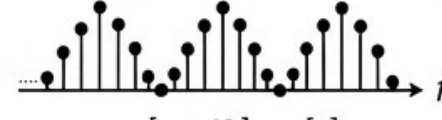
Fundamental period: T_0

Aperiodic CT signal $x(t)$



Discrete-Time (DT) Signals $x[n]$

Periodic DT signal $x[n]$



$$x[n + N_0] = x[n]$$

Fundamental period: $N_0 \in \mathbb{Z}^+$

Aperiodic DT signal $x[n]$



Fourier Series (FS)

$$x(t) = \sum_{k=-\infty}^{\infty} C_k e^{jk\omega_0 t}$$

$$C_k = \frac{1}{T_0} \int_{t_0}^{t_0+T_0} x(t) e^{-jk\omega_0 t} dt$$

$$\omega_0 = \frac{2\pi}{T_0}$$

(Discrete spectrum at $\omega = k\omega_0$)

Fourier Transform (FT)

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega$$

(Continuous spectrum)

Laplace Transform (LT)

$$X(s) = \int_{0^-}^{\infty} x(t) e^{-st} dt$$

$$s = \sigma + j\omega$$

(Generalises the Fourier transform for causal signals)

Sampling and Reconstruction

Sampling (continuous $\rightarrow$ discrete)

$$x[n] = x(nT_s)$$

$$T_s = \frac{1}{f_s}$$

Reconstruction (discrete $\rightarrow$ continuous)

If $x(t)$ is bandlimited to B Hz and $f_s > 2B$, then

$$x(t) = \sum_{n=-\infty}^{\infty} x[n] \text{sinc}\left(\frac{t - nT_s}{T_s}\right)$$

(Ideal sinc interpolation, perfect reconstruction)

$$X_F(\omega) = X_L(s) \Big|_{s=j\omega}$$

(if the $j\omega$ -axis is in the ROC)

Discrete-Time Fourier Series (DTFS)

$$x[n] = \sum_{k=0}^{N_0-1} C_k e^{j\frac{2\pi}{N_0} kn}$$

$$C_k = \frac{1}{N_0} \sum_{n=0}^{N_0-1} x[n] e^{-j\frac{2\pi}{N_0} kn}$$

$$\Omega_k = \frac{2\pi}{N_0} k$$

(Discrete spectrum at $\Omega = \Omega_k$)

Discrete-Time Fourier Transform (DTFT)

$$X(e^{j\Omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\Omega n}$$

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\Omega}) e^{j\Omega n} d\Omega$$

(Continuous spectrum, periodic in 2π)

Z-Transform (ZT)

$$X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n}$$

$$z = re^{j\Omega}$$

(Generates the DTFT)

$$X(e^{j\Omega}) = X(z) \Big|_{z=e^{j\Omega}}$$

(if the unit circle $|z| = 1$ is in the ROC)

Summary and Connection to ELEC2134

Periodic signals

CTFS and DTFS represent periodic signals using discrete frequency lines.

Aperiodic signals

CTFT and DTFT represent aperiodic signals using a continuous frequency variable.

Continuous-time relation

For causal/right-sided CT signals, the CTFT is obtained from the one-sided Laplace transform on $s = j\omega$ when the transform exists or provided the $j\omega$ -axis lies in the region of convergence.

Discrete-time relation

For DT signals, the DTFT is the Z-transform evaluated on the unit circle $z = \exp(j\Omega)$, provided $|z| = 1$ lies in the ROC.

Sampling connection

Sampling links CT and DT descriptions: $x[n] = x(nT_s)$. Exact reconstruction is possible for bandlimited $x(t)$ when $f_s > 2B$ using ideal sinc interpolation.

Unified viewpoint

All of these are related complex-exponential representations used to analyse signals and systems used for signal and system analysis.

ELEC2134 Scope

In ELEC2134, the assessable focus is the continuous-time branch: **Fourier Series, Fourier Transform and Laplace Transform**. DTFS, DTFT, Z-transform and sampling are included here only to show the bigger picture.

Where they are used

- **Fourier Series (FS)**: periodic continuous-time signals (e.g. steady-state analysis, power systems)
- **Fourier Transform (FT)**: aperiodic continuous-time signals (e.g. signals and systems, communications)
- **Laplace Transform (LT)**: continuous-time systems (e.g. transient analysis, stability, initial conditions)
- **Discrete-Time Fourier Series (DTFS)**: periodic discrete-time signals (e.g. digital signal processing)
- **Discrete-Time Fourier Transform (DTFT)**: aperiodic discrete-time signals (e.g. DSP, spectral analysis)
- **Z-Transform (ZT)**: discrete-time systems (e.g. digital filter design, stability, initial conditions)

ELEC2134: Circuits and Signals

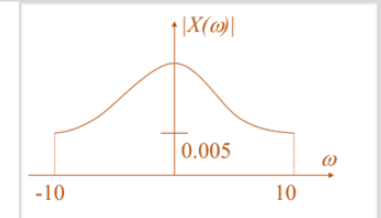
Transform Methods : Fourier Transforms Tutorial questions

Pass-Level Tutorial Questions (Fourier Transforms)

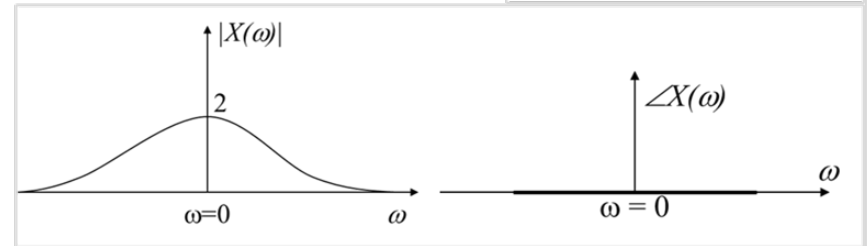
Question 1: Using the definition of the Fourier transform (FT), determine the frequency-domain representation $X(\omega)$ for the following signals. Sketch the corresponding magnitude spectra $|X(\omega)|$: (a) $x(t) = e^{-3t} u(t-1)$ (b) $x(t) = e^{-|t|}$

Ans:

$$(a) \quad X(\omega) = \frac{e^{-(j\omega+3)}}{j\omega+3}; \quad |X(\omega)| = \frac{e^{-3}}{\sqrt{\omega^2+9}}; \quad \varphi(\omega) = \angle X(\omega) = -\omega - \tan^{-1}\left(\frac{\omega}{3}\right)$$



$$(b) \quad X(\omega) = \frac{2}{\omega^2+1}; \quad |X(\omega)| = \frac{2}{\omega^2+1}; \quad \angle X(\omega) = 0$$

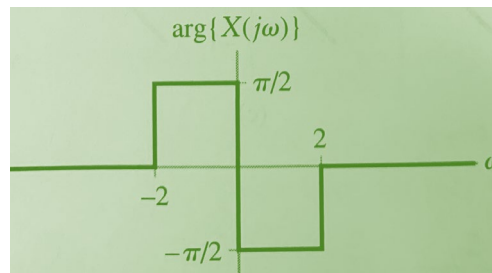
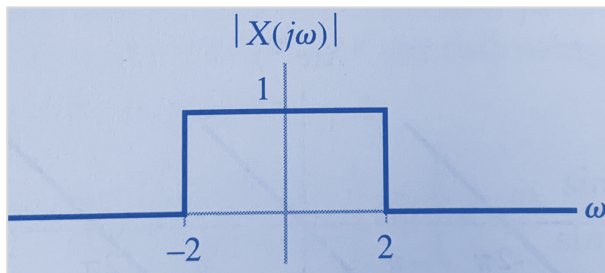


Question 2: Determine the time-domain signals $x(t)$ corresponding to the following Fourier transforms (FTs):

$$(a) \quad X(\omega) = e^{-2\omega} u(\omega); \quad (b) \quad X(\omega) = e^{-|\omega|}$$

$$\text{Ans: } (a) \quad x(t) = \frac{1}{2\pi(2-jt)}; \quad (b) \quad x(t) = \frac{2}{t^2+1}$$

Question 3: Determine the time-domain signal $x(t)$ corresponding to the following Fourier transform $X(\omega)$:

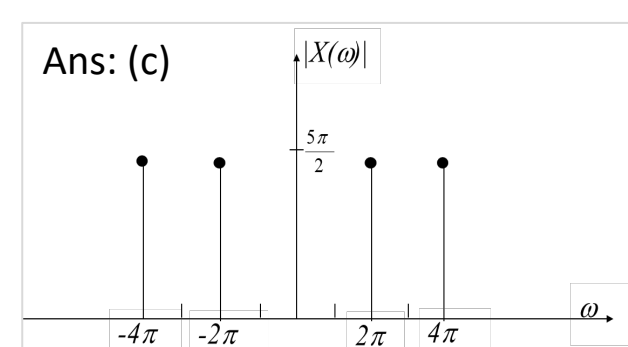
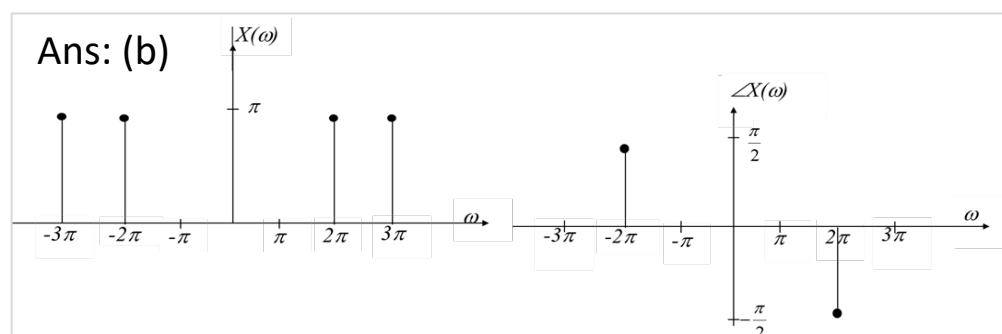
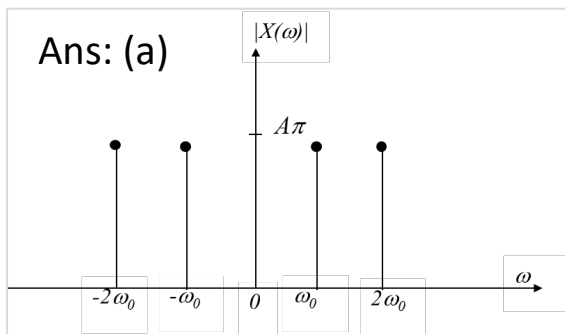


$$\text{Ans: } x(t) = \begin{cases} 0 & t = 0 \\ \frac{1 - \cos(2t)}{\pi t} & t \neq 0 \end{cases}$$

Pass-Level Tutorial Questions (Fourier Transforms)

Question 4:

- (a) Find the Fourier transform $X(\omega)$ of $x(t) = A\sin(\omega_0 t) + A\cos(2\omega_0 t)$ and sketch the magnitude spectrum $|X(\omega)|$.
- (b) For $x(t) = \sin(2\pi t) + \cos(3\pi t)$, sketch the magnitude $|X(\omega)|$ and phase $\angle X(\omega)$ spectra.
- (c) Consider the modulated signal $x(t) = 5\cos(3\pi t)\cos(\pi t)$; Determine $X(\omega)$ and sketch the amplitude (magnitude) spectrum $|X(\omega)|$.

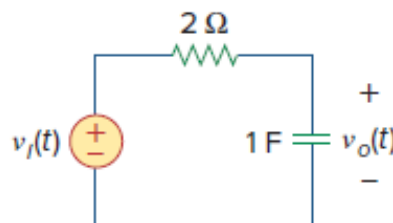


Question 5:

- (a) If $F(\omega) = \delta(\omega - 10) - 2\delta(\omega) - 3\delta(\omega + 30)$, sketch $|F(\omega)|$ and derive $f(t)$ using the definition of the inverse Fourier transform.
- (b) Determine the Fourier transform of the Fourier series $f(t)$ given below. Obtain the amplitude and phase spectra.

$$f(t) = \frac{1}{2} + \frac{2}{\pi} \sin(t) + \frac{2}{3\pi} \sin(3t) + \frac{2}{5\pi} \sin(5t)$$

Question 6: Find $v_o(t)$ in the given circuit if $v_i(t) = 2e^{-3t}u(t)$

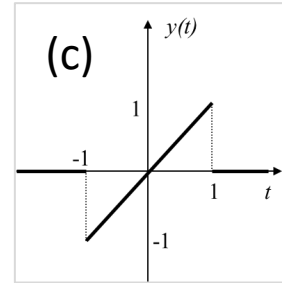
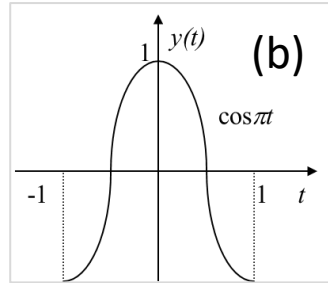
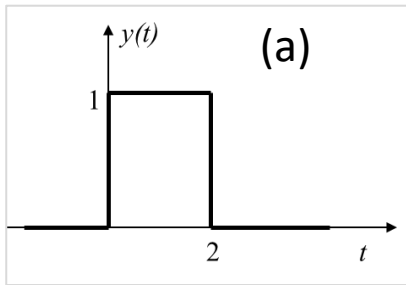


Ans:

$$v_o(t) = 0.4(e^{-0.5t} - e^{-3t})u(t)$$

Higher-Level Tutorial Questions (Fourier Transforms)

Question 1: Using the FT pair $x(t) = \begin{cases} 1 & |t| \leq 1 \\ 0 & \text{otherwise} \end{cases} \xleftrightarrow{FT} X(\omega) = \frac{2 \sin \omega}{\omega}$ and the FT properties, determine the frequency-domain representations for the signals in (a), (b), and (c) below.



Ans:

$$(a) \quad Y(\omega) = e^{-j\omega} \frac{2 \sin \omega}{\omega}$$

$$(b) \quad Y(\omega) = \frac{\sin(\omega - \pi)}{\omega - \pi} + \frac{\sin(\omega + \pi)}{\omega + \pi}$$

$$(c) \quad Y(\omega) = \frac{2j}{\omega} \left(\cos \omega - \frac{\sin \omega}{\omega} \right)$$

Question 2:

(a) Prove that the Fourier transform of a Gaussian is also a Gaussian. Given $x(t) = e^{-at^2}$, $a > 0$, use the FT definition to show that

$$X(\omega) = \sqrt{\frac{\pi}{a}} e^{-\frac{\omega^2}{4a}}.$$

(b) Explain the effect of increasing a on:

(i) the time-domain width of $x(t)$ (what happens to the time-domain width of $x(t)$ as the parameter a increases.)

(ii) the frequency-domain width of $X(\omega)$ (what happens to the frequency-domain width of $X(\omega)$ as a increases.)